

IILM UNIVERSITY

ARTIFICIAL INTELLIGENCE INTERNSHIP

Three-Project AI Portfolio

Rule-Based AI Chatbot • Data Classification Using AI • AI Recommendation Logic

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Program & Semester: B.Tech CSE (AI/ML), 2nd Year, 3rd Semester

Organization Name: DecodeLabs

Training Batch: 2026

Internship Overview

Three practical Artificial Intelligence projects completed under the DecodeLabs Industrial Training Kit.

Project 1: explicit rule-based decision making.

Project 2: supervised learning and classification.

Project 3: preference matching and recommendation logic.

Overall progression: Rules → Learning from Data → Personalization.

Project 1 — Rule-Based AI Chatbot

Goal: build a chatbot that responds to predefined user inputs.

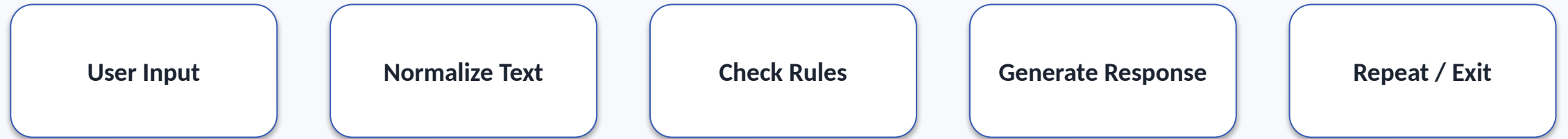
Uses if/elif/else decision logic and a continuous while loop.

Handles greetings, common questions, help requests, and exit commands.

Input is normalized using `strip()` and `lower()`.

Learning outcome: control flow, decision making, input handling, basic AI concepts.

Project 1 — Working Flow



Example: “hello” → greeting response

Example: “bye” → goodbye response and loop termination

Project 2 — Data Classification Using AI

Goal: build a basic classification model using a small dataset.

Dataset: Iris — 150 samples, 4 features, 3 classes.

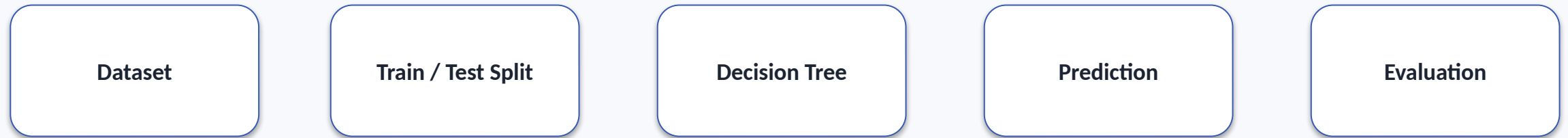
Workflow: load → inspect → split → train → predict → evaluate.

Model: DecisionTreeClassifier.

Split: 80% training and 20% testing, with stratification.

Evaluation: accuracy, classification report, and confusion matrix.

Project 2 — Supervised Learning Pipeline



The model learns patterns from labeled training examples and predicts classes for unseen test examples.

Project 3 — AI Recommendation Logic

Goal: create a simple recommendation system based on user preferences.

User enters interests such as action, sci-fi, thriller, drama, or comedy.

Each movie is represented by a set of genre attributes.

Jaccard similarity compares common attributes with total unique attributes.

Items are ranked by similarity and the top recommendations are displayed.

Project 3 — Similarity Example

User preferences = {action, sci-fi, thriller}

The Matrix = {action, sci-fi, thriller}

Intersection = 3

Union = 3

Jaccard Similarity = $3 / 3 = 1.00$

Result: The Matrix is a perfect match for these preferences.

Tools & Technologies

Python — core implementation language.

VS Code / Google Colab — development and execution.

pandas — dataset inspection and handling.

scikit-learn — supervised classification and evaluation.

Python sets and similarity logic — recommendation matching.

Skills & Learning Outcomes

Control flow and rule-based decision making.

Data handling and supervised learning basics.

Model training, prediction, and evaluation.

Pattern matching and similarity scoring.

Understanding the progression from programmed rules to data-driven and personalized AI.

Project Comparison

Aspect	Project 1	Project 2	Project 3
Approach	Rules	Supervised Learning	Similarity
Input	Text	Features	Interests
Output	Response	Class	Recommendations
Main Skill	Control Flow	Model Training	Pattern Matching

Challenges & Future Improvements

Expand chatbot vocabulary and add more conversational rules.

Compare multiple classification algorithms and test new data.

Add user ratings and weighted preferences to the recommendation system.

Increase dataset/item catalogue size.

Develop a simple graphical or web interface for the projects.

Conclusion

The internship provided a practical foundation in Artificial Intelligence.

Project 1 established explicit decision-making.

Project 2 introduced supervised machine learning.

Project 3 introduced personalized recommendation logic.

Together, the projects form a compact AI portfolio for further learning.

Thank You